

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Recalls the dimensions for energy and angular speed and forms an equation for dimensional consistency	AO1.1a	M1	$[E] = \text{ML}^2\text{T}^{-2}$ $[\omega] = \text{T}^{-1}$ $[I] = \frac{\text{ML}^2\text{T}^{-2}}{(\text{T}^{-1})^2} = \text{ML}^2$
	Completes a rigorous argument using both dimensions for energy and angular velocity to verify that the dimensions of I are ML^2	AO2.1	R1	
(b)	Uses dimensions to form a correct expression for the dimensions of $[I^\alpha W^\beta h^\gamma]$	AO1.1a	M1	$[I^\alpha W^\beta h^\gamma] = (\text{ML}^2)^\alpha (\text{MLT}^{-2})^\beta (\text{L})^\gamma$ $= \text{M}^{\alpha+\beta} \text{L}^{2\alpha+\beta+\gamma} \text{T}^{-2\beta}$ $\alpha + \beta = 0$ $-2\beta = 1$ $2\alpha + \beta + \gamma = 0$ $\alpha = 0.5$ $\beta = -0.5$ $\gamma = -0.5$
	Forms three simultaneous equations in three unknowns from 'their' $[I^\alpha W^\beta h^\gamma]$	AO1.1a	M1	
	Obtains correct values for α, β, γ CAO	AO1.1b	A1	
				Total 5 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Shows that $c = 0$ by considering the dimensions of mass and deduces that the speed does not depend on the density of the liquid	AO2.2a	M1	$LT^{-1} = (LT^{-2})^a \times L^b \times (MT^{-3})^c$ $0 = c$ Since $c = 0$, v does not depend on the density of the liquid. So David's model is incorrect.
	Rejects David's model because speed is shown not to depend on density	AO2.3	A1	
(b)	Uses dimensions to form an equation for dimensional consistency.	AO1.1a	M1	$LT^{-1} = (LT^{-2})^a \times L^b$ $1 = a + b$ $-1 = -2a$ $a = \frac{1}{2}$
	Obtains correct values for a and b .	AO1.1b	A1	

			$b = \frac{1}{2}$
Total 4 marks			

Q3.

$$T^1 = L^\alpha (MLT^{-2})^\beta (ML^{-1})^\gamma$$

M1A1

$$\alpha + \beta - \gamma = 0$$

$$\beta + \gamma = 0$$

Getting three equations

m1

$$-2\beta = 1$$

$$\beta = -\frac{1}{2}$$

$$\gamma = \frac{1}{2}$$

$$\alpha = 1$$

Solution

m1
A1F

[5]

Q4.

$$L = M^\alpha (LT^{-1})^\beta (LT^{-2})^\gamma$$

M1A1

$$\beta + \gamma = 1$$

$$-\beta - 2\gamma = 0$$

$$\alpha = 0$$

Getting three equations

m1

$$\gamma = -1$$

Solution

m1

$$\beta = 2$$

A1

5

[5]

Q5.

$$LT^{-1} = L^{\alpha} \times (ML^{-3})^{\beta} (LT^{-2})^{\gamma}$$

M1

There is no M on the left hand side,
so $\beta = 0$.

E1

$$LT^{-1} = L^{\alpha + \gamma} T^{-2\gamma}$$

Dependent on M1

m1

$$\alpha + \gamma = 1$$

$$-2\gamma = -1$$

Equating corresponding indices

m1

$$\gamma = \frac{1}{2}$$

A1

$$\alpha = \frac{1}{2}$$

A1

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Q6.

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(a) $MLT^{-2} = \frac{[G]MM}{L^2}$

M1A1

$$[G] = L^3 M^{-1} T^{-2}$$

A1F

3

(b) $t = km^{\alpha} R^{\beta} G^{\gamma}$
 $T = M^{\alpha} L^{\beta} M^{-\gamma} L^{3\gamma} T^{-2\gamma}$

L, M, T for G are needed to gain M1

M1A1F

$$-2\gamma = 1 \Rightarrow \gamma = -\frac{1}{2}$$

$$\alpha - \gamma = 0 \Rightarrow \alpha = -\frac{1}{2}$$

$$\beta + 3\gamma = 0 \Rightarrow \beta = \frac{3}{2}$$

Getting 3 equations

m1

Solution

m1

Finding α, β, γ

A1F

5

[8]

Q7.

(a) $T^1 = L^a \times M^b \times (LT^{-2})^c$

M1A1

There is no M on the left, so $b = 0$

E1

3

(b) $T^1 = L^{a+c} \times M^0 \times T^{-2}$

M1

$$\begin{cases} -2c = 1 \\ a + c = 0 \end{cases}$$

equating corresponding indices

m1

$$\alpha = \frac{1}{2}, c = -\frac{1}{2}$$

solution

m1

$$\therefore \text{Period} = k^{\frac{1}{3}} g^{-\frac{1}{3}}$$

constant needed

A1F

4

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Q8.

(a) (i) $ML^{-1}T^{-1}$

B1

1

(ii) $MLT^{-2}L^{-2} = ML^{-1}T^{-2}$
B2 for simplified form

B2,1

2

(b) $L^3 \times T^{-1} = L^{-1} \times L^a \times (MLT^{-2}L^{-2})^b \times (ML^{-1}T^{-1})^c$

M1A1F

$$= L^{-1} \times L^a \times M^b \times L^b \times T^{-2b} \times L^{-2b} \times M^c \times L^{-c} \times T^{-c}$$

$$= L^{-1+a-b-c} \times M^{b+c} \times T^{-2b-c}$$

PI

m1

$$\begin{cases} -1+a-b-c=3 \\ b+c=0 \\ -2b-c=-1 \end{cases}$$

Getting 3 equations

m1

Solution (finding at least one of a, b, c)

$$a = 4, b = 1, c = -1$$

M1

A1F

6

[9]

Q9.

(a) Dimension of a force is MLT^{-2}

B1

Dimension of $\frac{mM}{r^2}$ is M^2L^{-2}

Dimension of G is $\frac{MLT^{-2}}{M^2L^{-2}}$

M1

$$= M^{-1}L^3T^{-2}$$

A1

3

- (b) Inserting dimensions:
 $LT^{-1} = (M^{-1}L^3T^{-2})^\alpha M^\beta L^{-\gamma}$

M1

$$= M^{\beta-\alpha} L^{3\alpha-\gamma} T^{-2\alpha}$$

A1✓

Equating terms in T; $\alpha = \frac{1}{2}$

M1

Equating terms in M; $\beta = \frac{1}{2}$

Equating terms in L; $\gamma = \frac{1}{2}$

cao

A1

4

[7]

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Examiner reports

Q1.

Although students demonstrated a good understanding of dimensional notation, this question proved to be more challenging than expected, with a mean mark of 62% for part (a) and 59% for part (b).

In part (a) a number of students did not appear to know that $\frac{1}{2}$ was dimensionless. In addition, whilst many students knew the correct dimensions for energy, the dimensions of ω were often incorrectly stated.

In part (b), the majority of students were able to start with the correct line $[I^\alpha W^\beta h^\gamma] = (ML^2)^\alpha (MLT^{-2})^\beta (L)^\gamma$ but often did not then collect terms together correctly or subsequently failed to form three correct equations. The most successful students here always clearly stated their three equations before solving.

Q3.

Questions on dimensional analysis have been answered well in the past examination series. However, for this paper, a significant number of candidates used MLT^{-1} , instead of MLT^{-2} , for the dimensions of force. Also, some candidates used M or LM^{-1} , instead of ML^{-1} , for the dimensions of q . There were many candidates who made mistakes in collecting and simplifying the indices from their dimensional equation.

Q4.

The great majority of the candidates were able to answer the opening question of the paper correctly and gained full marks. The candidates seemed well versed in dimensional analysis. However, a small number of candidates used MLT^{-2} instead of LT^{-2} for the dimensions of g .

Q5.

This question was answered generally very well. The candidates were familiar with the concept of dimensional analysis. A relatively small number of candidates committed errors in equating the corresponding indices from their equations of dimensions. Occasionally, candidates made the mistake of replacing $(ML^{-3})^\beta$ with $M^\beta L^{-3}$.

Q6.

This question was answered well by the great majority of candidates. A small minority of the candidates seemed to be unfamiliar with the concept of dimensional analysis. Some candidates used L instead of L^2 for the dimensions of x^2 and a few candidates used LT^{-2} instead of MLT^{-2} for the dimensions of the force. Some candidates benefited from follow-through marks in part (b).

Q7.

The response to this question was quite variable. Too often, candidates were unable to answer this question correctly. Evidently, these candidates were not prepared for questions based on dimensional analysis. Some candidates' response to this question

was flawed as these candidates started with the formula $T = 2\pi \sqrt{\frac{1}{g}}$ and showed that $T = T$.

Q8.

This question was answered well by the great majority of the candidates. A small minority of the candidates seemed to be unfamiliar with the concept of dimensional analysis. Some candidates benefited from follow-through marks in part (b).

Q9.

Part (a) was well done; only a few candidates did not use MLT^{-2} as the dimension of force.

In part (b), most candidates knew the appropriate method but the manipulation of indices required was found to be challenging.



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